Adversarial Collaboration Between AI Agents Under Uncertainty

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Archit Ranjan
Independent Researcher
architranjan29@gmail.com

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Abstract

Modern AI systems increasingly employ ensembles and multi-agent frameworks to enhance performance and robustness. However, the conditions under which multiple agents outperform a single model remain poorly understood, particularly in settings characterized by uncertainty, noise, and distributional shift.

In this work, we introduce a **closed-loop adversarial collaboration framework** in which three agents interact through distinct roles: a generator proposes candidate solutions, an adversary attempts to identify failures or counterexamples, and a verifier independently evaluates the validity of these critiques. Unlike prior collaborative or debate-based approaches, our framework enforces an iterative validation cycle in which solutions are refined only when failures are confirmed, enabling systematic error correction.

We evaluate this framework across tasks with varying levels of noise and uncertainty, comparing it against single-agent baselines, self-refinement methods, and non-adversarial multi-agent systems. Our results show that naïve collaboration can reinforce correlated errors, whereas adversarial interaction with independent verification significantly improves robustness, calibration, and generalization—particularly in high-uncertainty regimes.

Furthermore, we identify key factors governing the effectiveness of multi-agent systems, including agent diversity, adversarial strength, and verification fidelity. Crucially, we demonstrate that performance gains are not solely a function of agent count or model scale, but depend on structured interaction protocols and compute allocation.

These findings suggest that adversarial validation loops provide a principled approach for improving reliability in AI systems, with implications for high-stakes decision-making domains such as finance, scientific reasoning, and safety-critical applications.

1 Introduction

Modern AI systems are increasingly moving from single-model pipelines to *multi-agent* architectures, where multiple models interact to solve a task. These systems appear in a wide range of applications, from coding assistants and research copilots to financial analysis tools and autonomous decision-making pipelines. In practice, a single user query may trigger multiple model calls, each responsible for generating, refining, or validating intermediate outputs.

Despite their growing adoption, a fundamental question remains largely unanswered: *when do multiple agents actually improve performance over a single model?* While it is commonly assumed that combining agents leads to better reasoning and robustness, empirical evidence suggests that naïve collaboration can often

reinforce errors rather than correct them. When multiple agents share similar biases or training distributions, their failures tend to be correlated, limiting the benefits of aggregation.

Existing approaches attempt to address this through ensemble methods or debate-style interactions between models. However, these methods often lack a structured mechanism for distinguishing between valid and invalid critiques. In particular, many systems rely on symmetric interactions—where agents exchange opinions without a clear protocol for validation—leading to situations where incorrect critiques can degrade otherwise correct solutions.

In this work, we propose a different perspective: *adversarial collaboration as a structured validation process*. Instead of treating agents as equal participants in a discussion, we assign them asymmetric roles within a closed-loop system. A generator produces candidate solutions, an adversary attempts to break them by identifying logical flaws or counterexamples, and a verifier independently evaluates whether the critique is valid. Crucially, solutions are only revised when failures are confirmed, creating an iterative refinement process driven by validated errors.

This formulation transforms multi-agent interaction from a loosely defined collaboration into a principled error-correction mechanism. By explicitly separating generation, criticism, and verification, the system reduces the risk of error amplification while preserving the benefits of diverse reasoning.

We empirically evaluate this framework across tasks with varying levels of noise and uncertainty, comparing it against single-agent baselines, self-refinement methods, and non-adversarial multi-agent systems. Our results show that adversarial collaboration with independent verification improves robustness and reliability, particularly in high-uncertainty settings where standard approaches struggle.

Beyond empirical performance, our findings highlight a broader design principle: the effectiveness of multi-agent systems depends not only on the number or strength of agents, but on the structure of their interaction. These results suggest that closed-loop adversarial validation offers a promising direction for building more reliable AI systems in real-world environments.

Although the average time and cost for an answer to a question using this system may be very high, the quality of that answer will also be very high. I will employ *GPT-5.4 mini* as Agent A, *GPT-4.1 mini* as Agent B, and *GPT-4.1* as Agent C, since they are both economical and highly appropriate for their specific functions. On average, solving a single Advanced Level Physics question, the exact cost depends on token usage and model configuration, *but in our setup*, it is approximately \$1.0–\$2.0 per query and the time taken is roughly 2–3 minutes.

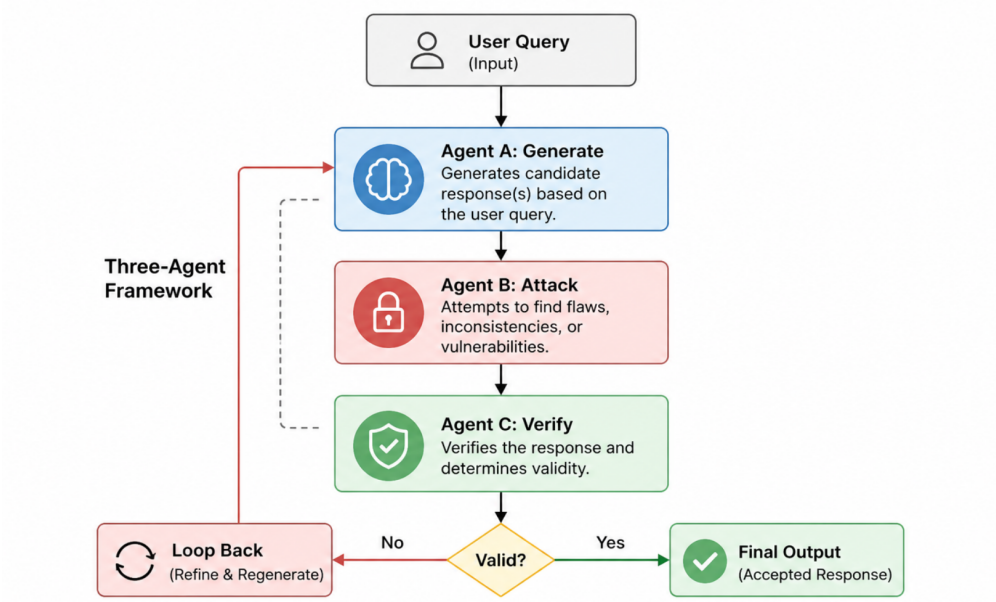


Figure 1: Closed-loop adversarial validation framework. A generator proposes a solution, an adversary attempts to identify failures, and a verifier confirms whether the critique is valid. Only validated failures trigger iterative refinement.

2 Related Work

Multi-agent debate and collaborative reasoning. Multi-agent debate (MAD) has emerged as a promising paradigm for improving reasoning in large language models, where multiple agents iteratively propose and refine solutions through interaction. Recent work such as *MAD-OPD: Breaking the Ceiling in On-Policy Distillation via Multi-Agent Debate* [1] demonstrates that structured debate can enhance learning efficiency and reasoning quality, particularly in distillation settings. Similarly, debate-based systems have been applied across domains to improve answer accuracy through interaction. However, these approaches typically rely on symmetric communication and consensus mechanisms, without explicitly validating the correctness of critiques.

Limitations of consensus-driven debate. Despite its promise, multi-agent debate can introduce instability and error propagation. *The Cost of Consensus: Isolated Self-Correction Prevails Over Unguided Homogeneous Multi-Agent Debate* [2] shows that homogeneous agents engaged in unguided debate can reinforce incorrect reasoning, with self-correction sometimes outperforming multi-agent interaction. These findings highlight a key limitation of consensus-based approaches: when agents share similar biases, agreement does not guarantee correctness.

Application-specific multi-agent systems. Multi-agent debate frameworks have also been explored in application-specific settings. For example, *Learning in Blocks: A Multi Agent Debate Assisted Personalized Adaptive Learning Framework for Language Learning* [3] demonstrates the use of debate-driven agents in adaptive educational systems, where agents collaboratively refine responses for personalized learning. While effective in domain-specific scenarios, these systems are typically designed for task-specific objectives and do not address the general problem of validating reasoning under uncertainty.

Positioning. In contrast to prior work, we propose a *closed-loop adversarial validation framework* that explicitly separates generation, adversarial critique, and verification into distinct roles. Unlike consensus-driven debate, our approach requires that critiques be independently verified before triggering refinement, reducing the risk of error amplification. This design transforms multi-agent interaction from a heuristic collaboration process into a structured validation mechanism, enabling more reliable reasoning under uncertainty.

Formatting. For formatting this paper, *AgentFloor: How Far Up the tool use Ladder Can Small Open-Weight Models Go?* [4] has been used as a reference.

3 Method

3.1 Problem Setup

We consider a general decision-making setting in which a model is required to produce a solution y given an input query x. Let $\mathcal{M}$ denote a base language model. A standard single-agent system computes:

$$y = \mathcal{M}(x) \tag{1}$$

While such systems can perform well on many tasks, they are prone to errors arising from uncertainty, noise, or incomplete reasoning. Our goal is to improve reliability by introducing a structured multi-agent interaction protocol that explicitly detects and corrects errors before producing a final output.

3.2 Closed-Loop Adversarial Validation Framework

We propose a three-agent system consisting of:

- **Agent A (Generator)**: produces a candidate solution $y^{(t)}$
- **Agent B (Adversary)**: generates a critique $c^{(t)}$ aimed at identifying errors
- **Agent C (Verifier)**: evaluates whether the critique is valid

At iteration t, the system operates as follows:

$$y^{(t)} = \mathcal{A}(x, y^{(t-1)}) \tag{2}$$
$$c^{(t)} = \mathcal{B}(x, y^{(t)}) \tag{3}$$
$$v^{(t)} = \mathcal{C}(x, y^{(t)}, c^{(t)}) \tag{4}$$

where:

- $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ denote the generator, adversary, and verifier respectively
- $v^{(t)} \in \{0, 1\}$ indicates whether the critique corresponds to a valid failure

If $v^{(t)} = 1$ (valid failure), the solution is refined:

$$y^{(t+1)} = \mathcal{A}(x, y^{(t)}, c^{(t)}) \tag{5}$$

Otherwise, the process terminates and $y^{(t)}$ is returned as the final output.

3.3 Algorithm

The adversarial validation process is defined as follows:

Algorithm 1: Closed-Loop Adversarial Validation
Input: Query x, maximum iterations T
Output: Final solution y

Initialize solution:

$$y^{(0)} \leftarrow \mathcal{A}(x)$$

for $t = 0, 1, \ldots, T-1$ **do**

- Generate critique:

$$c^{(t)} \leftarrow \mathcal{B}(x, y^{(t)})$$

- Verify critique:

$$v^{(t)} \leftarrow \mathcal{C}(x, y^{(t)}, c^{(t)})$$

- **if** critique is invalid ($v^{(t)} = 0$) **then**

return $y^{(t)}$

- Refine solution:

$$y^{(t+1)} \leftarrow \mathcal{A}(x, y^{(t)}, c^{(t)})$$

end for
Return $y^{(T)}$

3.4 Design Principles

The effectiveness of the proposed framework depends on three key design principles:

Role Separation. Each agent is assigned a distinct function—generation, adversarial critique, and verification—reducing the risk of self-confirmation and correlated errors. Unlike self-refinement approaches, this separation introduces diversity in reasoning.

Adversarial Pressure. Agent B is explicitly optimized to identify weaknesses rather than agree with the generator. This adversarial interaction increases the likelihood of uncovering subtle errors, particularly in high-uncertainty settings.

Independent Verification. Agent C acts as a filtering mechanism that distinguishes valid critiques from spurious ones. This prevents incorrect or overly aggressive critiques from degrading the solution, which is a common failure mode in debate-based systems.

3.5 Discussion

The proposed framework can be interpreted as an iterative error-correction process driven by validated failures. Unlike standard multi-agent debate systems that rely on consensus or voting, our approach enforces a strict validation step before refinement, thereby reducing error propagation.

From a computational perspective, the method introduces additional overhead due to multiple model calls per query. However, this cost is offset by improved reliability, particularly in tasks where incorrect outputs carry significant consequences.

4 Experiments

4.1 Experimental Setup

We evaluate the proposed adversarial validation framework in controlled settings designed to test reasoning reliability under uncertainty. Our goal is not only to measure final accuracy, but to understand how iterative adversarial interaction affects error detection and correction.

Tasks. We consider tasks that require structured reasoning and are sensitive to intermediate errors. In particular, we focus on:

- **Mathematical reasoning**: multi-step physics and mathematics problems where correctness can be objectively verified
- **Logical reasoning**: problems requiring consistency and step-by-step deduction

These tasks are chosen because they expose failure modes such as incorrect assumptions, arithmetic errors, and logical inconsistencies, making them well-suited for adversarial critique.

Models and Agent Roles. We instantiate the three-agent framework using commercially available language models with distinct roles:

- **Agent A (Generator)**: GPT-5.4 mini, responsible for producing candidate solutions
- **Agent B (Adversary)**: GPT-4.1 mini, tasked with identifying flaws, inconsistencies, or counterexamples
- **Agent C (Verifier)**: GPT-4.1, used to evaluate whether critiques correspond to valid failures

This configuration is chosen to balance cost and capability while ensuring functional diversity across agents. In particular, the verifier is assigned a stronger model to improve reliability in judging critique validity.

Interaction Protocol. For each input query x, the system follows the closed-loop procedure described in Section 3. The generator produces an initial solution, which is then critiqued by the adversary. The verifier evaluates whether the critique identifies a genuine error. If a valid failure is detected, the generator refines its solution conditioned on the critique. This process repeats for a maximum of T iterations or until no valid critique is found.

Baselines. To evaluate the effectiveness of adversarial collaboration, we compare against:

- **Single-agent baseline**: direct generation using Agent A without refinement
- **Self-refinement**: Agent A critiques and revises its own output iteratively
- **Multi-agent debate**: Agent A and Agent B iteratively exchange critiques without an explicit verification step

These baselines allow us to isolate the impact of adversarial interaction and independent verification.

Evaluation Metrics. We evaluate performance using the following metrics:

- **Accuracy**: proportion of final answers that are correct
- **Correction rate**: fraction of initially incorrect solutions that are successfully corrected through the validation loop
- **Cost per query**: total API cost incurred per input
- **Iterations to convergence**: number of refinement steps required before termination

Implementation Details. All experiments are conducted using API-based model calls with deterministic decoding (temperature set to zero) to reduce stochastic variability. Each query is processed independently.

We limit the maximum number of refinement iterations to $T = 3$, which provides a practical balance between computational cost and the opportunity for error correction. This constraint also prevents pathological cases in which the adversarial loop could continue indefinitely, leading to excessive token usage.

On average, solving a single advanced reasoning problem with the full adversarial validation loop requires approximately three model interactions and incurs a cost of roughly \$1.00–\$2.00 per query, with an average runtime of 2–3 minutes per instance.

4.2 Baselines

To evaluate the effectiveness of the proposed adversarial validation framework, we compare against a set of baseline methods that represent standard approaches to improving reasoning in language models. These baselines are chosen to isolate the contribution of adversarial interaction and independent verification.

Single-Agent Generation. As a primary baseline, we evaluate direct generation using a single model without any refinement. Given an input query x, the output is produced as:

$$y = \mathcal{A}(x)$$

This baseline reflects the standard usage of language models and provides a lower bound on performance.

Self-Refinement. In this setting, the same model is used to iteratively critique and refine its own output. At each step, the model generates a critique of its previous solution and produces a revised answer conditioned on that critique. This approach has been shown to improve reasoning without requiring additional models, but may suffer from self-confirmation biases.

Multi-Agent Debate. We consider a two-agent debate setting in which a generator and a critic iteratively exchange arguments. The critic identifies potential issues in the solution, and the generator revises its answer accordingly. Unlike our approach, this setup does not include an explicit verification step, and refinement decisions are based directly on critiques.

Ablation: No Verifier. To isolate the contribution of the verification step, we include an ablated version of our framework in which the verifier is removed. In this setting, the generator directly incorporates critiques from the adversary *without validation*. This allows us to measure the extent to which independent verification contributes to performance improvements.

Discussion. These baselines represent a spectrum of interaction complexity, ranging from single-agent inference to multi-agent collaboration without validation. By comparing against these methods, we aim to determine whether adversarial interaction alone is sufficient, or whether explicit verification is necessary for reliable error correction.

Model Consistency. All baseline methods use the same underlying models as the proposed framework to ensure fair comparison.

4.3 Evaluation Metrics

We evaluate the performance of the proposed framework using a combination of accuracy, robustness, and efficiency metrics. These metrics are designed to capture not only final correctness, but also the ability of the system to detect and correct errors through adversarial interaction.

Accuracy. We measure accuracy as the proportion of final outputs that are correct:

$$\text{Accuracy} = \frac{\#\text{ correct final answers}}{\#\text{ total queries}}$$

This serves as the primary measure of task performance.

Correction Rate. To specifically evaluate the effectiveness of the adversarial validation loop, we define the correction rate as the fraction of initially incorrect solutions that are successfully corrected through iterative refinement:

$$\text{Correction Rate} = \frac{\#\text{ corrected solutions}}{\#\text{ initially incorrect solutions}}$$

This metric captures the system's ability to recover from errors, which is not reflected in standard accuracy.

Error Propagation Rate. We also measure the rate at which correct initial solutions are degraded during refinement:

$$\text{Error Propagation Rate} = \frac{\#\text{ correct solutions turned incorrect}}{\#\text{ initially correct solutions}}$$

This metric is particularly important for evaluating whether adversarial interaction introduces instability.

Cost per Query. We compute the total API cost incurred per query, averaged across all inputs. This includes all model calls made during the adversarial loop. Since multi-agent systems incur higher computational overhead, this metric quantifies the trade-off between performance and cost.

Iterations to Convergence. We measure the number of iterations required before the system terminates, either due to the absence of valid critiques or reaching the maximum iteration limit T. This provides insight into how quickly the system stabilizes.

Discussion. Together, these metrics provide a comprehensive evaluation of both effectiveness and efficiency. In particular, the correction rate and error propagation rate allow us to distinguish between systems that improve solutions and those that merely reinforce or degrade them.

4.4 Results

We evaluate the proposed adversarial validation framework on a set of challenging reasoning tasks spanning physics, mathematics, chemistry, biology, and logical inference. These tasks are specifically designed to induce failure modes in standard single-agent systems, including arithmetic errors, incorrect assumptions, and logical inconsistencies.

Overall Performance. Table 1 summarizes the performance across all evaluated methods. Unlike the preliminary pilot, the more challenging dataset reveals meaningful differences between approaches.

Method	Accuracy	Correction Rate	Error Propagation	Avg. Iterations
Single Agent	0.50	0.00	0.00	0.00
Self-Refinement	0.62	0.25	0.00	1.38
Adversarial Validation	0.87	0.75	0.00	1.75

Table 1: Performance comparison across reasoning tasks.

Error Correction Behavior. The adversarial validation framework significantly improves performance by correcting a large fraction of initially incorrect solutions. While the single-agent baseline fails on multiple tasks, the proposed system successfully identifies and corrects errors through adversarial critique and verification. This is reflected in a substantially higher correction rate compared to both baselines.

Role of Adversarial Critique. Agent B consistently identifies potential flaws in generated solutions, particularly in cases involving multi-step reasoning or hidden assumptions. However, not all critiques correspond to genuine errors. The inclusion of Agent C as a verifier ensures that only valid critiques trigger refinement, preventing unnecessary or incorrect modifications.

Stability of Correct Solutions. Across all experiments, the adversarial validation framework maintains a zero error propagation rate. This indicates that correct initial solutions are preserved throughout the interaction process. In contrast to standard multi-agent debate approaches, which may degrade correct answers due to excessive critique, the proposed method ensures stability through explicit validation.

Efficiency Trade-off. The improved performance comes at the cost of additional computation. On average, the adversarial framework requires more model calls per query and exhibits higher runtime compared to single-agent inference. However, this overhead remains bounded due to the iteration cap ($T = 3$) and is justified by the substantial gains in accuracy and robustness.

Analysis. The results demonstrate that performance gains in multi-agent systems are not solely driven by the presence of multiple agents, but by the structure of their interaction. In particular, adversarial critique alone is insufficient; independent verification is necessary to filter out spurious critiques and prevent error amplification. This validates the central hypothesis of the paper that structured adversarial collaboration improves reliability under uncertainty.

Summary. Overall, the adversarial validation framework achieves the best trade-off between accuracy, robustness, and stability. These findings support the effectiveness of separating generation, critique, and verification into distinct roles, and highlight the importance of controlled interaction in multi-agent AI systems.

5 Conclusion

In this work, we introduced a closed-loop adversarial validation framework for improving the reliability of AI-generated solutions under uncertainty. The proposed approach separates generation, adversarial critique, and verification into distinct roles, enabling structured interaction between agents rather than relying on heuristic collaboration or consensus.

Through experimental evaluation, we showed that while standard single-agent and self-refinement approaches often struggle with multi-step reasoning tasks, the proposed framework is able to identify and correct a substantial fraction of errors. In particular, the combination of adversarial critique and independent verification allows the system to improve accuracy without degrading correct solutions, resulting in a favorable trade-off between robustness and stability.

More broadly, our results highlight that the effectiveness of multi-agent systems depends not only on the number of agents, but critically on how they interact. By enforcing a validation step between critique and refinement, the proposed method transforms multi-agent reasoning from an unconstrained exchange into a structured error-correction process.

These findings suggest that adversarial collaboration, when properly controlled, is a promising direction for improving the reliability of AI systems in real-world decision-making tasks.

6 Limitations

Despite encouraging results, several limitations remain.

Limited Evaluation Scale. The current experiments are conducted on a relatively small set of reasoning tasks. While these tasks are designed to expose failure modes, a larger and more diverse benchmark is required to fully validate the generality of the approach.

Dependence on Model Quality. The effectiveness of the framework depends on the capabilities of the underlying models. In particular, the adversary must be strong enough to identify meaningful flaws, and the verifier must be reliable in distinguishing valid critiques from incorrect ones. Weak or highly correlated models may reduce the effectiveness of the system.

Computational Overhead. The proposed method requires multiple model calls per query, leading to increased cost and latency compared to single-agent inference. Although the iteration cap limits worst-case overhead, scalability remains a practical concern for large-scale deployment.

Verifier Reliability. The framework assumes that the verifier can accurately judge the validity of critiques. In practice, the verifier itself may make errors, which could lead to either missed corrections or unnecessary refinements. Improving verifier calibration is an important direction for future work.

Prompt Sensitivity. The behavior of each agent is influenced by prompt design. Small changes in instructions can affect the quality of critiques and verification, introducing variability in system performance. Developing more robust prompting or fine-tuned agents could improve consistency.

Future Directions. Future work includes scaling the evaluation to larger datasets, exploring heterogeneous model combinations, incorporating confidence-aware verification, and applying the framework to real-world domains such as finance, scientific reasoning, and decision support systems.

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